

MARIE HANSEN

DATA SCIENCE ENGINEER

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<https://github.com/mn-coding-cop>

SUMMARY

Data scientist and AI/ML engineer with six years of experience across the Finance, Retail, Healthcare, and Marketing industries. Proficient in AI, Deep Learning, Machine Learning, and Computer Vision, with a proven track record in developing and deploying innovative analytics solutions. Adept at translating complex data into actionable insights, driving strategic decision-making, and optimizing performance. Skilled in leading cross-functional teams and committed to staying at the forefront of technology to deliver exceptional results. Passionate about leveraging data to solve real-world problems and enhance business outcomes.

TECHNICAL SKILLS

- **Data Science Concepts:** Mathematics, statistical analysis, modeling & inferencing techniques, data visualization, data manipulation, data exploration
- **Data Science Skills and Tools:** Pandas, Numpy, Scipy, Matplotlib, Seaborn, Plotly, Tableau, MixPanel, Google Looker
- **AI/ML Concepts:** Support Vector Machines (SVM), Decision Trees, Random Forests, Gradient Boosting, Reinforcement Learning, Deep learning architectures, Neural Networks
- **AI/ML Skills and Tools:** TensorFlow, PyTorch, scikit-learn, Keras, Natural Language Processing (NLP), Large Language Models (LLM), BERT, LaMDA, GPT-2/3/4, LLaMa, Huggingface, LangChain, AutoGPT, AgentGPT, OpenAI API, AWS, Azure, GCP, MLflow, Kubeflow
- **Data Engineering Concepts:** Data pipeline, workflow management, data manipulation and automation, ETL processes, data modeling, relational/non-relational databases
- **Data Engineering Skills and Tools:** Airflow, Hadoop, Spark, AWS Glue, data bricks, AWS Redshift, Snowflake, SQL Server, ETL (AWS), Amazon S3, SQL, NoSQL

PROFESSIONAL EXPERIENCE

Data Science Engineer, TheBitSoft

2020 - Present

- Led a cross-functional team of 6 full-stack web developers and 2 data scientists in the development of diverse software projects.
- Led creating a recommendation system using collaborative filtering techniques, resulting in a 20% boost in user engagement.
- Spearheaded the implementation of advanced data science tools and methodologies for DoD and Government policymakers, enhancing decision-making processes in international relations.
- Guided the team in setting and achieving goals aligned with stakeholder requirements, ensuring accountability and delivery of high-quality products.
- Developed and optimized models such as RAG and Vector DB using LLM, QLoRa, Docker, and AWS Lambda for scalability within a serverless architecture.
- Maintained 95.7% reliability using Google SRE standards while facilitating rapid feature development.
- Integrated various data sources through API services, Apache Beam, and BigQuery, centralizing data for efficient analysis.
- Analyzed web and mobile usage to prioritize intents for chatbot creation, increasing user engagement to serve over 700 users daily.
- Implemented advanced AI models, including CNNs, RNNs, LSTMs, and Transformers with Word Embeddings, achieving a 19% improvement in accuracy and processing efficiency for over 1.3 million words.
- Utilized Python libraries (NumPy, Pandas, SciPy, Matplotlib, Plotly, FeatureTools) for robust data analytics, cleaning, and feature engineering, ensuring high data quality.
- Enhanced chatbot performance by analyzing Watson Assistant data, resulting in improved response accuracy for over 2000 daily queries.

Machine Learning Engineer, Confiz**2019 - 2020**

- Developed a custom data analytics dashboard for the telecommunications industry, showcasing business insights and predictive models.
- Created a Tensor Flow deep learning model for customer sentiment analysis, improving accuracy by 15%.
- Implemented data retrieval from Hadoop HBase on Spark using PySpark for a NoSQL database.
- Deployed operationalized models to a RESTful API using Python Flask and Docker on AWS.
- Collaborated across teams to conduct A/B tests, enhancing user engagement by 15%.
- Applied NLP techniques such as sentiment analysis and text classification to enhance product recommendations, resulting in a 15% sales increase.
- Presented strategic insights to C-suite and stakeholders based on data analysis.
- Developed supplementary materials for enhanced student success, including conceptual exercises, additional references, practice tests, and impromptu video lectures upon student request.
- Educated students in Data Engineering, Data Science, and Machine Learning disciplines.
- Provided direct guidance to new students entering Lambda's data science program.
- Evaluated assignments and delivered constructive feedback to inspire and guide students effectively.
- Built scalable data processing pipelines with Apache Spark, optimizing real-time model inference and handling over 100000 transactions per day.
- Used AWS Glue, lambda and SageMaker to build and deploy AI applications cutting operational costs by 12% and speeding up by 18%.
- Deployed visual analytics dashboards with Tableau and ggplot to provide actionable insights into customer demographics and sales trends, resulting in a 15% improvement.

Software Developer, BrainX**2018 - 2019**

- Developed predictive models using machine learning resulting in 35% improvement in prediction speed
- Led the implementation of Deep Learning algorithms, achieving a 97% success rate in image recognition tasks
- Automated time series forecasting models, resulting in a more accurate prediction and saving an average of 20 hours a week
- Formulated A/B test plans for ML models, driving key decisions with a 15% error reduction.
- Contributed to AI/ML research initiatives, collaborating with cross-functional teams to translate business requirements into impactful solutions.
- Optimized machine learning algorithms for streamlined handling of large datasets and real-time processing.
- Built a retail chatbot from scratch with Python, Dialogflow, and Flask, focusing on intent detection and entity recognition to boost customer engagement.
- Developed and delivered machine learning training, improving team's ML adoption rate by 20%

EDUCATION

Masters from George Mason University